

THE CALABI–YAU-TO-CHIRAL FUNCTOR Φ AND THE UNIVERSAL TRACE IDENTITY

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ABSTRACT. SCAFFOLD STUB. The Calabi–Yau-to-chiral functor $\Phi: \text{CY}_d \rightarrow \text{Chiral-Alg}$ promotes a compact Calabi–Yau d -fold X to a chiral (factorisation) algebra $\mathcal{A}_X$ on an auxiliary curve. The principal invariant preserved is $\kappa_{\text{ch}}(\mathcal{A}_X) = \sum_q (-1)^q h^{\circ,q}(X)$ (Hodge supertrace, discharged with algebraic-vs-geometric model discipline). The output E_n -scope stratifies by dimension: $\Phi(X) \in E_2$ -chiral for $d \leq 2$ and $\Phi(X) \in E_1$ -chiral for $d \geq 3$. The universal trace identity unifying Vol I $K = -c_{\text{ghost}}$ (BRST) with $\kappa_{\text{BKM}}(\Phi_N) = c_N(\circ)/2$ is the bridge from Vol I’s universal conductor to Vol III’s Borchers weight. Full content to be drafted from the chapter inputs below.

SCOPE AND STATUS

Primary theorems advertised (status per Vol III CLAUDE.md theorem table, 2026-04-17):

- Φ preserves Koszul duality on the Koszul locus: .
- $\kappa_{\text{ch}} = \chi_{\text{Hodge}}$ (Hodge supertrace): (thm:kappa-hodge-supertrace-identification).
- E_n -output scope $n = 2$ for $d \leq 2$, $n = 1$ for $d \geq 3$: (FM43 scope guard).
- Universal trace identity $K_{\text{univ}} = -c_{\text{ghost}}$: (Vol I chap:universal-conductor).

CHAPTER INPUTS (POINTERS)

The body of this standalone is to be assembled from the following Vol III chapter files (no content duplication; \input pointers only):

```
\input{../chapters/theory/cy_to_chiral.tex}
\input{../chapters/theory/quantum_chiral_algebras.tex}
\input{../chapters/theory/en_factorization.tex}
\input{../chapters/examples/cy_d_kappa_stratification.tex}
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CROSS-VOLUME DEPENDENCIES

- Vol I chap:universal-conductor (universal trace identity $K_{\text{univ}} = -c_{\text{ghost}}$).
- Vol I thm:A-infinity-2 (properad bar-cobar; Vol III Φ is a corollary of Francis–Gaitsgory (cor:chiral-KK-formal-smoothness)).
- Vol II thm:chiral-higher-deligne (target category for $\Phi(X)$ ’s derived centre at $d = 3$).